

Kunihiro Asada

professor

基本信息

姓名： Kunihiro Asada

所属院校： 东京大学

地区： Japan

年龄范围： 50岁以上

学科领域

集成电路设计与集成系统

微电子科学与工程

研究领域

数字集成电路设计

模拟与射频IC设计

射频集成电路

低功耗电路与能量管理

混合信号集成技术

微电子器件与制造

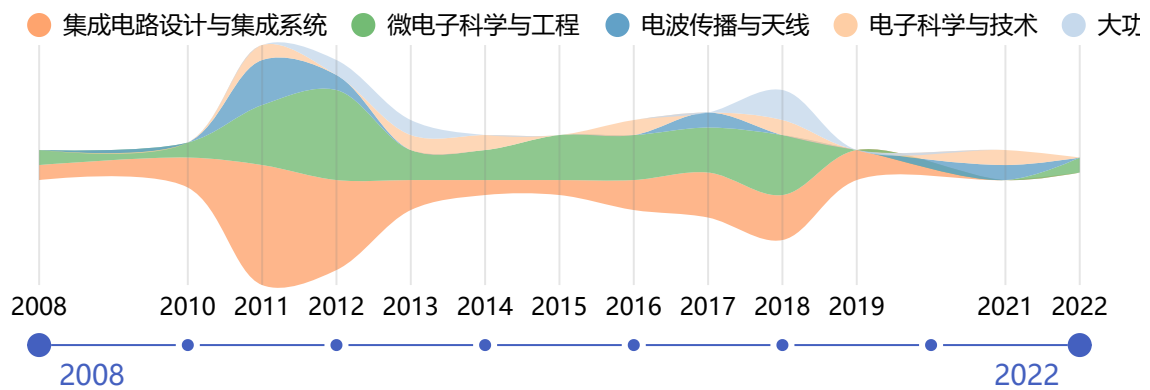
超大规模集成电路设计

EDA工具与电路验证

个人简介

Kunihiro Asada, a professor with a doctorate, is affiliated with the VLSI Design and Education Center at the University of Tokyo. His address is 7 - 3 - 1 Hongo, Bunkyo - ku, Tokyo 113, JAPAN. His specialties include VLSI Circuits and System Design, and Semiconductor Devices. He developed Bode for Windows, a linear circuit simulator for Windows. You can reach him via email at asada(a)silicon.u - tokyo.ac.jp, fax at +81 - 3 - 5841 - 8911, or phone at +81 - 3 - 5841 - 6671.

近年发表的论文



4-Cycle-Start-Up Reference-Clock-Less Digital CDR Utilizing TDC-Based Initial Frequency Error Detection with Frequency Tracking Loop

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Chin, Meikan; Nakura, Toru; Asada, Kunihiro

被引用次数: 0

2022

Shock-wave Transceiver Integration for Mm-wave Active Sensing Applications

2021 INTERNATIONAL CONFERENCE ON IC DESIGN AND TECHNOLOGY (ICICDT)

Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2021

Spatial resolution improvement for point light source detection in scintillator cube using SPAD array with multi pinholes

IEICE ELECTRONICS EXPRESS

Iizuka, Tetsuya; Xu, Kai; Yang, Xiao; Nakura, Toru; Asada, Kunihiro

被引用次数: 1

2019

A 16-bit 2.0-ps Resolution Two-Step TDC in 0.18- μ m CMOS Utilizing Pulse-Shrinking Fine Stage With Built-In Coarse Gain Calibration

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Enomoto, Ryuichi; Iizuka, Tetsuya; Koga, Takehisa; Nakura, Toru; Asada, Kunihiro

被引用次数: 21

2019

Fault Detection of VLSI Power Supply Network Based on Current Estimation From Surface Magnetic Field

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Takahashi, Daigo; Iizuka, Tetsuya; Nguyen Ngoc Mai-Khanh; Nakura, Toru; Asada, Kunihiro

被引用次数: 13

2019

Optimal Design Method of Sub-Ranging ADC Based on Stochastic Comparator

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Hossain, Md Maruf; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 2

2018

Triangular Active Charge Injection Method for Resonant Power Supply Noise Reduction

IEICE TRANSACTIONS ON ELECTRONICS

Kano, Masahiro; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2018

Quick-Start Pulse Width Controlled PLL with Frequency and Phase Presetting

IEICE TRANSACTIONS ON ELECTRONICS

Nakura, Toru; Kagaya, Tsukasa; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 1

2018

Time-domain approach for analog circuits in deep sub-micron LSI

IEICE ELECTRONICS EXPRESS

Asada, Kunihiro; Nakura, Toru; Iizuka, Tetsuya; Ikeda, Makoto

被引用次数: 17

2018

Digitally-Controlled Compensation Current Injection to ATE Power Supply for Emulation of Customer Environment

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Terao, Naoki; Nakura, Toru; Ishida, Masahiro; Ikeno, Rimon; Kusaka, Takashi; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2018

Analysis and design of impulse signal generator based on current-mode excitation and transmission line resonator

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Kanjanavirojkul, Parit; Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 0

2018

Noninvasive Localization of IGBT Faults by High-Sensitivity Magnetic Probe With RF Stimulation

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Nguyen Ngoc Mai-Khanh; Nakajima, Shigeru; Iizuka, Tetsuya; Mita, Yoshio; Asada, Kunihiro

被引用次数: 25

2018

A Gate Delay Mismatch Tolerant Time-Mode Analog Accumulator Using a Delay Line Ring

IEICE TRANSACTIONS ON ELECTRONICS

Yano, Tomohiko; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2017

A CMOS Broadband Transceiver with On-Chip Antenna Array and Built-In Pulse-Delay Calibration for Millimeter-Wave Imaging Applications

IEICE TRANSACTIONS ON ELECTRONICS

Nguyen Ngoc Mai-Khanh; Asada, Kunihiro

被引用次数: 3

2017

A PLL Compiler from Specification to GDSII

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 1

2017

Design, Analysis and Implementation of Pulse Generator by CMOS Flipped on Glass for Low Power UWB-IR

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Kanjanavirojkul, Parit; Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 1

2017

Power Supply Voltage Control for Eliminating Overkills and Underkills in Delay Fault Testing

IEICE TRANSACTIONS ON ELECTRONICS

Ishida, Masahiro; Nakura, Toru; Kusaka, Takashi; Komatsu, Satoshi; Asada, Kunihiro

被引用次数: 0

2016

A 15 x 15 single photon avalanche diode sensor featuring breakdown pixels extraction architecture for efficient data readout

JAPANESE JOURNAL OF APPLIED PHYSICS

Yang, Xiao; Zhu, Hongbo; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 4

2016

An On-Chip Measurement of PLL Transfer Function and Lock Range through Fully Digital Interface

IEICE TRANSACTIONS ON ELECTRONICS

Kikkawa, Toshiyuki; Nakura, Toru; Asada, Kunihiro

被引用次数: 0

2016

Dynamic Power Integrity Control of ATE for Eliminating Overkills and Underkills in Device Testing

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Ishida, Masahiro; Nakura, Toru; Kusaka, Takashi; Komatsu, Satoshi; Asada, Kunihiro

被引用次数: 0

2016

An Asynchronous Summation Circuit for Noise Filtering in Single Photon Avalanche Diode Sensors

JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS

Yang, Xiao; Zhu, Hongbo; Nakura, Toru; Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 0

2016

Resonant Power Supply Noise Reduction by STO Capacitors Fabricated on Interposer

IEICE TRANSACTIONS ON ELECTRONICS

Nakura, Toru; Kano, Masahiro; Yoshizawa, Masamitsu; Hattori, Atsunori; Asada, Kunihiro

被引用次数: 1

2015

A Near-Field Magnetic Sensing System With High-Spatial Resolution and Application for Security of Cryptographic LSIs

IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT

Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Sasaki, Akihiko; Yamada, Makoto; Morita, Osamu; Asada, Kunihiro

被引用次数: 16

2015

Spacial Resolution Enhancement for Integrated Magnetic Probe by Two-Step Removal of Si-Substrate Beneath the Coil

IEEE TRANSACTIONS ON MAGNETICS

Mai-Khanh, Nguyen Ngoc; Iizuka, Tetsuya; Nakajima, Shigeru; Asada, Kunihiro

被引用次数: 6

2015

Comparative study of RF energy harvesting rectifiers and proposal of output voltage universal curves for design guideline

IEICE ELECTRONICS EXPRESS

Nakura, Toru; Matsui, Hiroaki; Asada, Kunihiro

被引用次数: 2

2015

A Structured Routing Architecture for Practical Application of Character Projection Method in Electron-Beam Direct Writing

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Ikeno, Rimon; Maruyama, Takashi; Komatsu, Satoshi; Iizuka, Tetsuya; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 3

2014

A Flash TDC with 2.6-4.2 ps Resolution Using a Group of Unbalanced CMOS Arbiters

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Komatsuta, Satoshi; Yamaguchi, Takahiro J.; Abbas, Mohamed; Nguyen Ngoc Mai Khanh; Tandon, James; Asada, Kunihiro

被引用次数: 2

2014

On-Chip Switched Decoupling Capacitor for Fast Voltage Hopping of DVS Systems

IEICE TRANSACTIONS ON ELECTRONICS

Kim, Jinmyoung; Nakura, Toru; Ishibashi, Koichiro; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 0

2013

High-Throughput Electron Beam Direct Writing of VIA Layers by Character Projection with One-Dimensional VIA Characters

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Ikeno, Rimon; Maruyama, Takashi; Komatsu, Satoshi; Iizuka, Tetsuya; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 1

2013

An Integrated High-Precision Probe System in 0.18- μ m CMOS for Near-Field Magnetic Measurements on Cryptographic LSIs

IEEE SENSORS JOURNAL

Nguyen Ngoc Mai-Khanh; Iizuka, Tetsuya; Yamada, Makoto; Morita, Osamu; Asada, Kunihiro

被引用次数: 14

2013

Self Synchronous Circuits for Robust Operation in Low Voltage and Soft Error Prone Environments

IEICE TRANSACTIONS ON ELECTRONICS

Devlin, Benjamin; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 0

2013

Low Pass Filter-Less Pulse Width Controlled PLL Using Time to Soft Thermometer Code Converter

IEICE TRANSACTIONS ON ELECTRONICS

Nakura, Toru; Asada, Kunihiro

被引用次数: 7

2012

A 580 fs-Resolution Time-to-Digital Converter Utilizing Differential Pulse-Shrinking Buffer Ring in 0.18 μm CMOS Technology

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Miura, Satoshi; Yamamoto, Ryota; Chiba, Yutaka; Kubo, Shunichi; Asada, Kunihiro

被引用次数: 6

2012

On-Chip Switched Parasitic Capacitors of Sleep Blocks for Resonant Supply Noise Reduction

IEICE TRANSACTIONS ON ELECTRONICS

Kim, Jinmyoung; Nakura, Toru; Takata, Hidehiro; Ishibashi, Koichiro; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 0

2012

All-Digital Wireless Transceiver with Sub-Sampling Demodulation and Burst-Error Correction

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Bushnaq, Sanad; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 0

2012

All-Digital PMOS and NMOS Process Variability Monitor Utilizing Shared Buffer Ring and Ring Oscillator

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 8

2012

Energy Minimum Operation with Self Synchronous Gate-Level Autonomous Power Gating and Voltage Scaling

IEICE TRANSACTIONS ON ELECTRONICS

Devlin, Benjamin; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 3

2012

Frequency Resolution Enhancement for Digitally-Controlled Oscillator Based on a Single-Period Switching Scheme

IEICE TRANSACTIONS ON ELECTRONICS

Kodama, Kazutoshi; Iizuka, Tetsuya; Nakura, Toru; Asada, Kunihiro

被引用次数: 1

2012

A millimeter-wave resistor-less pulse generator with a new dipole-patch antenna in 65-nm CMOS

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Khanh, Nguyen Ngoc Mai; Sasaki, Masahiro; Asada, Kunihiro

被引用次数: 2

2012

All-digital tunable power amplifier consuming 0.03 mW/MHz using 0.18 μ m CMOS

IEICE ELECTRONICS EXPRESS

Bushnaq, Sanad; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 0

2012

A High Speed IC Truly Random Number Generator Based on Chaotic Sampling of Regular Waveform

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Ergun, Salih; Guler, Ulkuhan; Asada, Kunihiro

被引用次数: 52

2011

Variable Length Coded Address Compression for High-Speed 3-D Range-Finder Using Light-Section Method

IEICE TRANSACTIONS ON ELECTRONICS

Mandai, Shingo; Momma, Taihei; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 0

2011

Cascaded Time Difference Amplifier with Differential Logic Delay Cell

IEICE TRANSACTIONS ON ELECTRONICS

Mandai, Shingo; Nakura, Toru; Iizuka, Tetsuya; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 1

2011

A 0.18- μ m CMOS X-Band Shock Wave Generator with an On-Chip Dipole Antenna and a Digitally Programmable Delay Circuit for Pulse Beam-Formability

IEICE TRANSACTIONS ON ELECTRONICS

Nguyen Ngoc Mai Khanh; Sasaki, Masahiro; Asada, Kunihiro

被引用次数: 0

2011

On-Chip Resonant Supply Noise Canceller Utilizing Parasitic Capacitance of Sleep Blocks for Power Mode Switch

IEICE TRANSACTIONS ON ELECTRONICS

Kim, Jinmyoung; Nakura, Toru; Takata, Hidehiro; Ishibashi, Koichiro; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 1

2011

All-Digital On-Chip Monitor for PMOS and NMOS Process Variability Utilizing Buffer Ring with Pulse Counter

IEICE TRANSACTIONS ON ELECTRONICS

Iizuka, Tetsuya; Jeong, Jaehyun; Nakura, Toru; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 3

2011

1.0 Ps Resolution Time-to-Digital Converter Based-On Cascaded Time-Difference-Amplifier Utilizing Differential Logic Delay Cells

IEICE TRANSACTIONS ON ELECTRONICS

Mandai, Shingo; Iizuka, Tetsuya; Nakura, Toru; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 2

2011

A 0.25- μ m Si-Ge Fully Integrated Pulse Transmitter with On-Chip Loop Antenna Array towards Beam-Formability for Millimeter-Wave Active Imaging

IEICE TRANSACTIONS ON ELECTRONICS

Nguyen Ngoc Mai Khanh; Sasaki, Masahiro; Asada, Kunihiro

被引用次数: 1

2011

All-digital ramp waveform generator for two-step single-slope ADC

IEICE ELECTRONICS EXPRESS

Iizuka, Tetsuya; Asada, Kunihiro

被引用次数: 1

2011

A 65-nm CMOS Fully Integrated Shock-Wave Antenna Array with On-Chip Jitter and Pulse-Delay Adjustment for Millimeter-Wave Active Imaging Application

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Nguyen Ngoc Mai Khanh; Sasaki, Masahiro; Asada, Kunihiro

被引用次数: 2

2011

A 65 nm Gate-Level Pipelined Self-Synchronous FPGA for High Performance and Variation Robust Operation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Devlin, Benjamin; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 8

2011

A 8bit two stage time-to-digital converter using time difference amplifier

IEICE ELECTRONICS EXPRESS

Mandai, Shingo; Nakura, Toru; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 2

2010

Time Difference Amplifier with Robust Gain Using Closed-Loop Control

IEICE TRANSACTIONS ON ELECTRONICS

Nakura, Toru; Mandai, Shingo; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 3

2010

A Low Power and High Throughput Self Synchronous FPGA Using 65 nm CMOS with Throughput Optimization by Pipeline Alignment

IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES

Devlin, Benjamin Stefan; Nakura, Tort; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 52010

Dual Imager Core Chip with 24.8 Rangemaps/s 3-D and 58 fps 2-D Simultaneous Capture Capability

IEICE TRANSACTIONS ON ELECTRONICS

Mandai, Shingo; Nakura, Toru; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 02009

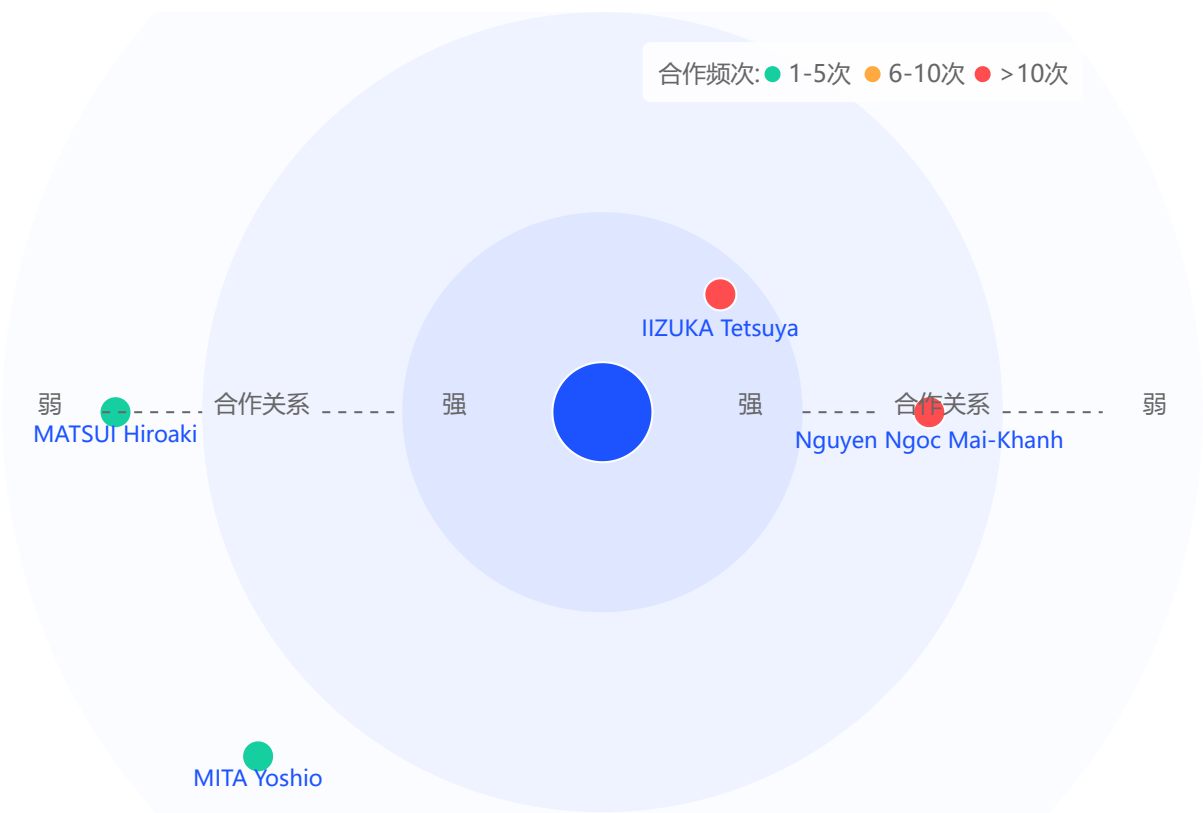
A temperature sensor with an inaccuracy of-1/+0.8°C using 90-nm 1-V CMOS for online thermal monitoring of VLSI circuits

IEEE TRANSACTIONS ON SEMICONDUCTOR MANUFACTURING

Sasaki, Masahiro; Ikeda, Makoto; Asada, Kunihiro

被引用次数: 672008

学术合作



IIZUKA Tetsuya 合作次数: 25
东京大学

Nguyen Ngoc Mai-Khanh 合作次数: 13
东京大学

MITA Yoshio

东京大学

合作次数: 1

MATSUI Hiroaki

东京大学

合作次数: 1
